SentinelRL: Budgeted Network Defence with Deep RL

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github.com/rohitkumarmourya-maker/SentinelRL

Abstract. SentinelRL allocates scarce defensive interventions in a synthetic network with stochastic attack propagation and noisy alerts. An exact public-history belief filter feeds a learned dueling Double DQN; a fixed temporal ensemble averages ten checkpoints from five training runs. The environment rewards critical-asset protection while penalizing damage, delayed confirmation and unnecessary isolation. On 10,000 paired test scenarios, 54 of 55 Bonferroni-adjusted return contrasts favor the ensemble; the tuned diagnostic rule remains an unresolved rival with a better point estimate. The claims concern this fixed deployment and prespecified test, with diagnosis and service trade-offs assessed separately.

1. Task and stochastic environment

A defender has $B_0 = 6$ response tokens over $T = 48$ steps. An intervention changes future exposure, observations and remaining resources, making the task sequential rather than a static alert-classification problem. Six hosts have fixed attributes

$$v = (.40, .68, .58, .82, .76, .48),$$
$$c = (.45, .62, .80, 1, .52, .96),$$

with $c_i \geq .80$ defining critical assets. Undirected edges are $(0,1)$, $(1,2)$, $(2,3)$, $(1,4)$, $(4,2)$, $(3,5)$ and $(2,5)$.

Initially one hidden foothold is at host 0 with probability .62, otherwise host 4. A compromised, unisolated source attempts to infect each susceptible, unisolated neighbour j with probability

$$q_j = .23 v_j (.20)^{p_j} (.25)^{d_j},$$

where p_j, d_j mark patch/decoy deployment. Sources are snapshotted: new infections spread on the next step. Isolation blocks incoming/outgoing movement; patch and decoy reduce risk without curing infection. Nonisolated hosts emit binary alerts with sensitivity .55 and false-positive rate .08. Independent pre-sampled noise supports paired comparisons. No real network traffic or adaptive attacker is modeled.

2. State, observation and causal belief

The augmented reward-relevant state has 56 dynamic entries: per host, compromise, patch, isolation, decoy, alert, alert EWMA, last inspection result/time and ever-confirmed flag, plus budget/time. The hidden compromise pattern $z \in \{0,1\}^6$ has 64 possibilities. Graph, v, c are constants.

Public observation $o_t \in [0,1]^{56}$ has nine entries per host (alert, EWMA, last inspection result, normalized inspection age, v, c, patch, isolation, decoy), plus $B_t/6, t/48$. Results encode unknown/clean/compromised as $0/.5/1$ and remain stale after hidden infection; EWMA weight is .35. No true infection count or actual entry identity is exposed.

Using only public action/alert/inspection history h_t, maintain $b_t(x) = P(z_t = x \mid h_t)$ over 64 patterns [1]. For an explicit inspection of i returning e, let $g_t(x) = \mathbf{1}[x_i = e]$; otherwise $g_t = 1$. Then

$$b_{t+1}(x') \propto \Omega(y_{t+1} \mid x') \sum_x P_{a_t}(x' \mid x) b_t(x) g_t(x).$$

Inspection conditions the pre-spread state; action-updated public defenses determine the transition, followed by the new alarm likelihood. Isolated sensors are ignored. Reset conditions the $.62/.38$ prior on initial alerts. Exactness assumes the known synthetic law. The Q input $\xi_t \in [0,1]^{126}$ contains public/history features 56, joint posterior 64 and marginals 6. Raw alert/EWMA/age slots are zeroed; the inspection-result slot becomes a public-history ever-positive bit. The posterior retains the evidence.

3. Actions and reward

The 25 actions are WAIT and six host-specific instances each of INSPECT, PATCH, ISOLATE and DECOY. A valid intervention costs one token. INSPECT reveals current compromise before spread. A public legality mask removes unavailable actions; WAIT is always legal.

Let u_t indicate a valid intervention, q_t an invalid action, f_t a first successful explicit confirmation, N_t newly infected hosts, U_t unisolated infected hosts, F_t a clean host newly isolated and L_t all clean isolated hosts. With ever-confirmed flag $v_{t,j}$, define $H_t = U_t \cap \{j : v_{t,j} = 0\}$ and $K_t = \{j \in U_t : c_j \geq .80\}$. Host sets use the post-action/post-spread state; the terminal clock is resulting $t = T$:

$$\begin{aligned} r_t = &-.16u_t - .45q_t + .40f_t \\ &- \sum_{j \in N_t} (2 + 4c_j) - .16 \sum_{j \in U_t} c_j \\ &- \sum_{j \in F_t} (.65 + .85c_j) - .03 \sum_{j \in L_t} c_j \\ &- .04 \sum_{j \in H_t} c_j - 1.4 \cdot \mathbf{1}[t = T] |K_t|. \end{aligned}$$

The objective is $\max_\pi E_\pi[\sum_{t=1}^T r_t]$. Confirmation pays once per host; preventive and containment actions earn their value through avoided future losses. Hidden truth calculates simulator rewards/metrics, not controller features.

4. Learned control and temporal ensemble

Double DQN separates online action selection from target evaluation [2]; dueling heads separate state value and action advantages [3]. Joint posterior 64 embeds to 32 ReLU units. Shared host heads take 50 inputs (9 host features, one marginal, two globals, six adjacency entries, 32 joint features), pass through 128/128 ReLU layers, and output four advantages. Mean host embedding (128), globals (2) and joint embedding (32) feed separate 162–128–1 WAIT/value heads. Centering the 25 advantages gives $Q = V + A - \text{mean}_a A$.

For up to three retained decision transitions, terminal flag d and next legal set $\mathscr{A}'$, $\gamma = 1$ gives

$$a^* = \arg\max_{a \in \mathscr{A}'} Q_\theta(\xi', a),$$
$$y = \sum_{j=0}^{k-1} R_{t+j} + (1-d) Q_{\bar{\theta}}(\xi', a^*).$$

MSE minimizes $E[(Q_\theta(\xi, a) - y)^2]$. Each R_i sums the actual rewards in its retained transition. Actual forced-WAIT suffix rewards fold into the preceding transition only when the public mask offers WAIT alone; all 48 environment steps execute. Duration-aware targets preserve the return exactly ($\gamma = 1$).

Five fresh seeds $S = \{17, 29, 43, 71, 89\}$ each retain fixed episode checkpoints $E = \{10000, 20000\}$:

$$Q_{\text{ens}}(\xi, a) = \frac{1}{10} \sum_{s \in S} \sum_{e \in E} Q_{s,e}(\xi, a).$$

Deployment takes legal argmax of this uniform mean. It is ONE fixed ensemble from five training runs, not ten independent replicas. No reward decoding, teacher, hand-added Q prior or strategic shield chooses its actions. Control is approximate; the filter's exactness does not imply optimal control.

5. Training and evaluation protocol

Architecture, loss and aggregation were selected on 200 development scenarios $600000 + 101i$ using development seeds 101/202. All five fresh 20k-episode runs and both fixed checkpoint times were retained without final-test selection. They execute 4.8 million environment steps and 1,719,035 optimizer updates. Settings are Adam 3×10^{-4}, batch 128, replay 30,000, warm-up 1,000 replay entries, episode-end updates, target copy every 500 updates and gradient-norm clip 10. Epsilon falls $1 \to .02$ over 240,000 actual steps. Logged training totals 63.4 worker-minutes on CPU (five parallel workers, one Torch thread each).

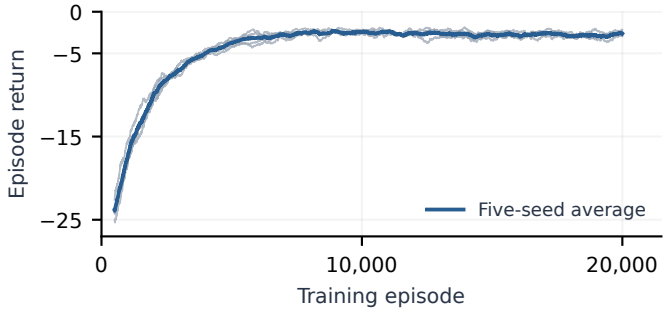


Fig. 1. Actual exploration-policy training returns: 500-episode moving means for five independent runs (gray), with their average (blue). These are descriptive traces, not confidence intervals or final ensemble performance.

Final testing uses 10,000 fresh scenarios $900000000 + 103i$, $i = 0, \ldots, 9999$, disjoint from known training/development/history. Configuration, checkpoints, comparison family and sample size froze before access. Shared exogenous noise pairs all policies by scenario. The 55 primary contrasts cover 22 comparator methods plus 33 fixed baseline checkpoints. A 50,000-resample bootstrap shares scenario draws while independently resampling actual comparator seed axes; the ensemble has one fixed-policy row. Bonferroni intervals use $\alpha = .05/55$. Secondary intervals use 2,000 resamples and are exploratory.

6. Primary return and trade-offs

The deployment obtains mean return -1.637 (nominal 95% CI $[-1.725, -1.552]$). The strongest learned comparator, the final-checkpoint graph ensemble (-2.174), has paired deficit 0.537 (simultaneous CI $[0.429, 0.646]$). The matched temporal graph ensemble gives -2.213; the paired gain is 0.576 ($[0.459, 0.695]$). Fifty-four contrasts have positive adjusted lower bounds.

The exception is the tuned diagnostic rule (-1.557): ensemble minus rule is -0.081 (adjusted CI $[-0.223, 0.063]$). This comparison is inconclusive; universal superiority and equivalence are not established. The five individual final models average -2.112 with seed SD 0.213; they are separate from the deployed ensemble.

Uncontained hosts average 0.0252 per episode. Explicit-confirmation coverage is only 8.62% (nominal CI $[8.10, 9.18]$%), versus 88.53% for the alert rule. Censored confirmation delay is 43.90 versus 5.57 steps. Low successful confirmation is a substantive weakness despite the return gains. The tuned rule has fewer uncontained hosts (0.0097) but more clean-host isolations (0.2633 versus 0.1656); these are exploratory mean comparisons.

Raw episodes, all family/fixed-checkpoint contrasts, per-seed metrics, source hashes and frozen manifests accompany the code. No legacy score from a different reward is used as a numerical benchmark.

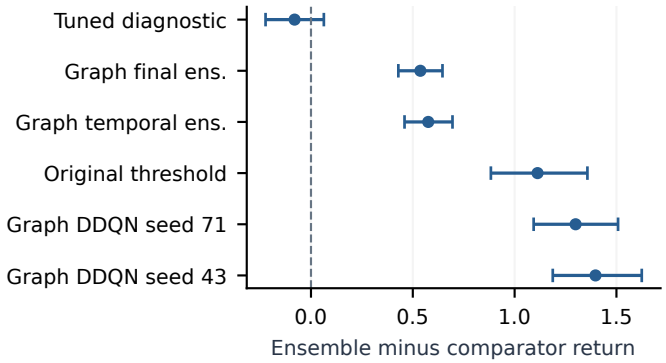


Fig. 2. The six most limiting simultaneous return-difference bounds in the prespecified family. Points are ensemble minus comparator mean return; bars are Bonferroni-adjusted bootstrap intervals over all 55 primary contrasts, using 50,000 resamples and 10,000 shared scenarios. The ensemble's trained members are fixed.

Policy	Return	Crit.	False iso.	Conf.%
Temporal DDQN	−1.637	0.0633	0.1656	8.62
Tuned diagnostic	−1.557	0.0729	0.2633	8.80
Graph final ens.	−2.174	0.0610	0.2110	9.84
Original threshold	−2.750	0.1085	0.0418	30.98
Original transfer	−4.960	0.2696	0.0190	92.65
Alert rule	−11.527	0.6043	0.0000	88.53

Table 1. Final means. Critical hosts ever compromised and clean-host isolations are counts/episode. Confirmation is the mean within-episode fraction of infected hosts explicitly diagnosed. Secondary intervals are exploratory; complete estimates accompany the raw results.

7. Fair comparison and scope

The 22 comparator methods include previous corrected RL agents, original rules and tabular encoding, matched 20k-episode graph-dueling retraining (learned-only and prior/shield hybrid), temporal/final ensembles and a diagnostic threshold rule tuned over 40 validation configurations. Original checkpoints are clearly marked domain transfers: their lawful 58D adapter uses causal posterior marginals and public inspection history, with the unavailable privileged new-infection-count channel fixed to 0. Transfer and retraining are distinct.

For infected host i, censored confirmation delay is $\min(t_i^{\text{confirm}}, T) - t_i^{\text{infection}}$, with ∞ for unconfirmed hosts. Isolated but unconfirmed infections remain censored, so containment need not imply diagnosis. Coverage accompanies this measure. Controls with zero false isolation preclude a strict win on every metric.

The known synthetic model and 2^N-state filter limit scalability and external validity. The shared head has restricted cross-host deployment context; several training/representation changes occur together, so results do not isolate one component's causal effect. Only one temporal ensemble is deployed: uncertainty is conditional on its fixed members, not variability across independently trained ensembles. Results establish a bounded comparison, not universal dominance or real-network effectiveness.

References

[1] L. P. Kaelbling, M. L. Littman and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence* 101:99–134, 1998. Author full text.

[2] H. van Hasselt, A. Guez and D. Silver. Deep reinforcement learning with Double Q-learning. *AAAI*, 2016. arXiv:1509.06461.

[3] Z. Wang et al. Dueling network architectures for deep reinforcement learning. *ICML*, PMLR 48:1995–2003, 2016. Publisher.